

SOP-GUARD PRO

Project Status & Launch Readiness Report

Prepared by	Date	Version	Audience
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EXECUTIVE SUMMARY

SOP-Guard Pro is an industrial SaaS compliance platform for SOP management and Preventive Maintenance. Development began mid-February 2026. The product has gone through a full v1 build, a diagnosed architectural overhaul, and a complete v2 rebuild. As of May 2026, all core features of the application are implemented and functional. The app is live on Vercel. The next critical step is a commissioner review meeting to validate compliance flows before moving into the commercial launch phase targeted for June 2026.

1. Project Background

SOP-Guard Pro is an industrial-grade SaaS compliance platform designed to centralize SOP (Standard Operating Procedure) management and Preventive Maintenance tracking for manufacturing and industrial organizations. The platform provides a full QA governance layer, digital audit trail, and real-time compliance notifications.

Development began in mid-February 2026 under ElkanHub. The product was built from scratch in two full iterations — Version 1 (completed and demoed) and a complete architectural rebuild as Version 2, which is the current live codebase.

The app is currently live on Vercel. One meeting has been held with the commissioner. The upcoming meeting is the second, and it is the critical gate before commercial launch preparations can begin.

2. Development Timeline

Period	Milestone	What Happened
Mid-Feb 2026	Project Start	Research phase begins. Domain knowledge gathering on SOP management, industrial compliance standards, and system architecture options.
Feb-Mar 2026	V1 Build	Version 1 built, tested, and demoed. Core flows functional: approval workflow, Change Controls, Pulse notifications, digital signatures. Demonstrated a working prototype end-to-end.
Mar 2026	V1 Diagnosis	8 critical architectural flaws identified in V1 — primarily the tree-node department model, fragmented notification system, and non-composable role logic. Decision made to rebuild rather than patch.
Mar-May 2026	V2 Rebuild	Full architectural rebuild on star-centralized model. Flat identity model (role + department tag + is_admin flag). Unified pulse_items table. 18 DB tables, all RLS-secured. All 13 planned feature phases implemented.
May 2026	Current Status	App is live on Vercel. All core feature modules complete. Marketing/landing pages added with placeholder pricing. Awaiting Commissioner Meeting #2 for compliance flow validation before commercial prep begins.

June 2026	Target Launch	Commercial launch target. Requires: compliance validation, pricing decisions, payment/subscription system, feature-plan segmentation, legal documents, and expanded AI features.
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3. Why We Built Version 2

Version 1 was built, tested, and demoed successfully. The approval flow ran end-to-end. However, as the system was put under real scrutiny, eight architectural flaws were identified that could not be patched without compounding complexity. The decision was made to rebuild rather than patch.

Key Flaws Diagnosed in V1

- Tree-node department architecture — SOPs could only belong to one department, creating walls that real industrial organizations do not have.
- QA visibility was a hardcoded architectural bypass, not a clean design decision.
- Adding a new department required a developer and a meaningful development task.
- Three separate notification systems (notices, approval alerts, PM alerts) doing the same job.
- Admin role was ambiguous — could not express 'Admin who is also a Manager.'

What V2 Fixed

- Star-centralized architecture: SOP Library and Equipment Registry as two central hubs. No department pages.
- Flat identity model: role + department tag + is_admin boolean. Fully composable.
- QA visibility is a natural outcome of the identity model, not a special case.
- Unified pulse_items table replacing three fragmented notification systems.
- Departments are now a text attribute — adding one is a single row insert.

4. Current Feature Status

The table below documents the build status of every feature module in SOP-Guard Pro as of May 2026.

Feature / Module	Status	Notes
— CORE APPLICATION —		
Authentication (Login/Signup/OAuth)	COMPLETE	Email + Google OAuth. Secure session management via Supabase Auth.
First-Run Setup Wizard	COMPLETE	Admin account initialization on first deployment.
User Onboarding (4-step wizard)	COMPLETE	Profile setup, department assignment, role, signature capture.
Role & Permission System	COMPLETE	Two roles (manager/employee) + is_admin flag. Composable. RLS-enforced at DB level.
Department Management	COMPLETE	Flat reference model. Add/remove departments as single row inserts.
User Management (Admin)	COMPLETE	Grant/revoke admin, deactivate users, admin password re-auth for sensitive ops.
— SOP LIBRARY —These features are COMPLETE but needs review...		
SOP Library (table view)	COMPLETE	TanStack Table, status filters, department scoping. Two-layer visibility rule applied.
SOP Viewer (.docx rendering)	COMPLETE	Microsoft Office Online Viewer via signed Supabase URLs. Pixel-perfect .docx display.
Two-Layer Visibility (own dept + org search)	COMPLETE	Employees see own dept SOPs by default; org-wide search available read-only.
SOP Upload & Submission	COMPLETE	Secure upload route, metadata capture, notes to QA. Submission creates QA Pulse alert.
Cross-Department SOP Tagging	COMPLETE	secondary_departments array. SOPs visible to all tagged departments.
SOP Tab Strip	COMPLETE	Zustand-powered. Up to 8 open SOPs. Note: tab state is memory-only, cleared on refresh.
SOP Acknowledgement Flow	COMPLETE	Employee acknowledgement tracked per version. Manager can view acknowledgement log.
Version History	COMPLETE	Immutable version chain. History viewable. One-click rollback (promote) not yet built.
— QA APPROVALS & CHANGE CONTROL —		
QA Approval Queue	COMPLETE	Pending SOPs and equipment submissions routed to QA Managers. Approve / Request Changes flow.
Approval Comment Thread	COMPLETE	QA-submitter dialogue thread. Re-submission loop functional.

Change Control (Significant changes)	COMPLETE	Auto-triggered on significant SOP update. Diff viewer, signature grid, completion trigger.
Minor Change Instant Activation	COMPLETE	Minor updates bypass Change Control. QA approval alone promotes SOP to Active.
Diff Viewer	COMPLETE	Two-column old vs new. Changed paragraphs highlighted using diff-match-patch.
Gemini AI Delta Summary	COMPLETE	3-5 bullet AI summary of document changes. Disclaimer shown. Google Gemini Flash.
Digital Signature Workflow	COMPLETE	Signature image captured at onboarding. Shown at signing. IP + timestamp logged to certificate.
Signatory Snapshotting	COMPLETE	Signatories locked at CC issuance. Subsequent role changes do not affect the existing CC.
Signature Waiver	COMPLETE	Admin can waive a signatory (documented in audit log).
QA Self-Approval Block	COMPLETE	QA Manager who submitted a SOP cannot approve their own submission.
SOP Locking During Active CC	COMPLETE	SOP locked=true while Change Control is pending. Prevents concurrent edits.
— EQUIPMENT REGISTRY & PM PLANNER —		
Equipment Registry	COMPLETE	Flat registry. Asset submission, QA approval, status lifecycle (pending → active → inactive).
Equipment Detail Page + QR Code	COMPLETE	Full asset detail view with service history and auto-generated QR code.
PM Task Generation (auto)	COMPLETE	First PM task auto-created on equipment approval. Next task created on completion.
PM Completion Logging	COMPLETE	Log completion with notes (required). Photo upload optional. Service history updated.
PM Task Reassignment	COMPLETE	Manager can reassign PM tasks to another team member.
Overdue PM Escalation	COMPLETE	Overdue CCs > 3 days auto-escalate to QA Manager via Pulse.
— PULSE & MESSAGING —		
The Pulse (real-time notification feed)	COMPLETE	Single pulse_items table. One Supabase Realtime subscription. Type-driven renderer.
Pulse Sound Notifications	COMPLETE	Different sounds for notices vs messages. Toggleable in Settings.
Notice / Broadcast System	COMPLETE	Send to individual, department, or everyone. Acknowledgement tracking.
Messages Module (direct + group)	COMPLETE	Full messaging with conversations, members, edit (15-min window), delete, read tracking.
— OTHER MODULES —		

Company Calendar	COMPLETE	PM dates auto-populated. Manual events (public or department). Built-in, not Google-linked.
Reports & Audit Log	COMPLETE	5 report types. CSV export. Append-only audit log. AI Risk Insights (Gemini).
Global Search	COMPLETE	TopNav search across the application.
Documentation Site	COMPLETE	MDX-powered docs at /docs. Dynamic slug routing. Public-facing.
— COMMERCIAL READINESS (NOT YET BUILT) —		
Landing Page & Marketing Pages	PARTIAL	Pages exist with design and copy. Pricing sections contain placeholder values only.
Pricing & Plan Decisions	PENDING	No pricing agreed. Requires commissioner input before pages can go live.
Feature-Plan Segmentation	NOT BUILT	All features currently accessible as one. No Starter/Pro/Enterprise gating implemented.
Payment Gateway	NOT BUILT	No Stripe, Paystack, or equivalent integrated.
Subscription / Billing System	NOT BUILT	No subscription lifecycle management built.
Email Marketing System	NOT BUILT	No email automation, drip sequences, or newsletter tooling set up.
AI Add-On Package (expanded AI features)	NOT BUILT	Current AI: delta summaries + risk insights. Additional AI features for paid tier TBD.
Legal / Privacy / Policy Documents	NOT BUILT	No Terms of Service, Privacy Policy, or SaaS agreement documents prepared.
SOP Version Rollback (Promote to Current)	NOT BUILT	History is viewable. One-click rollback button not yet implemented.
Custom Metadata Fields (Dynamic fields)	NOT BUILT	Planned (Phase 15) but not present in DB or UI. field_definitions table not created.

5. Outstanding Items Before Launch

The core application is feature-complete. The following items are not yet implemented and represent the gap between the current state and a commercially launchable product.

5.1 Pricing & Plan Structure

The marketing pages exist but contain placeholder pricing. No plan tiers have been defined (e.g., Starter, Professional, Enterprise). No feature gating has been implemented — all features are currently accessible as a single undivided product. Pricing decisions must come from the commissioner meeting.

5.2 Payment & Subscription System

No payment gateway (Stripe, Paystack, or equivalent) has been integrated. No subscription lifecycle management exists. This is a pre-requisite for any paying customer.

5.3 AI Add-On Package

The current AI implementation includes: Gemini-powered Change Control delta summaries and AI Risk Insights in the Reports module. The plan includes an AI add-on tier with expanded features. The specific additional features for this tier have not yet been defined and require agreement at the commissioner meeting.

5.4 Legal & Policy Documents

No legal documents have been prepared. The following are required before public launch:

- Terms of Service
- Privacy Policy
- SaaS / Subscription Agreement
- Data Processing Agreement (if handling data on behalf of clients)

5.5 Email Marketing

No email automation, drip sequences, or marketing newsletter system has been set up. This is not a blocker for launch but is needed for user acquisition and retention.

6. Known Issues & Technical Notes

Issue	Severity	Detail
Signed URL expiry during document viewing	LOW	If a user leaves the SOP viewer open for an extended session and the Supabase signed URL expires, the Office Viewer iframe may show an error. User must refresh. Does not affect data integrity.
Zustand tab strip is memory-only	LOW	The SOP tab strip (open documents) is stored in memory. A page refresh clears open tabs. No data is lost — SOPs are not affected. Minor UX friction only.
Service role required for signature upload at signup	RESOLVED	Previously required service role client for initial signature insertion during onboarding because the profile RLS policy blocked the operation before the profile existed. This has been addressed.
Custom metadata fields not implemented	NOTED	The dynamic field_definitions table (Phase 15) was planned but is absent from both the database and the UI. This is a post-launch feature. Does not affect core compliance functionality.

7. Technology Stack

SOP-Guard Pro is built on a modern, production-grade stack. All infrastructure is cloud-hosted with no on-premise dependencies.

Layer	Technology	Purpose
Framework	Next.js 15 (App Router) + TypeScript	Full-stack React. Server components for data fetching.
Database	Supabase (PostgreSQL + RLS)	18 tables. All RLS-secured. Identity enforced at DB layer.
Auth	Supabase Auth	Sessions, JWT, Google OAuth, email confirmation.
Realtime	Supabase Realtime	Powers The Pulse. Single subscription per user.
Storage	Supabase Storage	SOP .docx files, signatures, equipment photos. Private buckets.
AI	Google Gemini Flash	Change Control delta summaries. Audit log risk insights.
Styling	Tailwind CSS + Shadcn/UI	Utility-first. Brand colors: Navy #0D2B55 · Blue #1A5EA8 · Teal #00C2A8.
Hosting	Vercel	Edge deployment. Cron jobs for PM alerts and overdue checks.

8. Proposed Agenda — Commissioner Meeting #2

The following items are proposed for the upcoming meeting. This is the critical gate meeting before commercial launch work begins. All commercial decisions are pending the outcomes of this session.

#	Agenda Item	Why It Matters
1	Live walkthrough of all major compliance flows	Commissioner needs to see the SOP submission → QA approval → Change Control → signature chain in action. This validates whether the implemented process matches expected regulatory standards.
2	Verify the QA governance model	Confirm that QA as sole approval authority, the two-layer visibility rule, and the digital signature certificate approach align with the compliance standards the commissioner expects.
3	Pricing structure and plan tiers	Pricing decisions must be finalized before the marketing pages can go live. This includes plan names, feature tiers (Starter / Pro / Enterprise / AI Add-On), and target price points.
4	AI Add-On scope definition	The AI package is planned as a premium add-on. The commissioner and developer need to agree on what specific AI features will be included beyond the current delta summary and risk insights.

5	Launch timeline & next steps	June launch is the current target. Agreement is needed on what must be complete before launch vs. what can be phased in post-launch (e.g., email marketing, advanced AI features).
6	Legal and policy documents	Terms of Service, Privacy Policy, and SaaS agreement are not yet drafted. A decision is needed on whether to engage a lawyer or use template-based documents before launch.

9. Launch Readiness Plan — Path to June 2026

The following phased plan outlines the work required between now and the June 2026 target launch date. Phases are listed in dependency order.

Phase	Tasks	Dependency / Blocker
Phase A Meeting #2	Commissioner walkthrough Flow validation Pricing decisions AI scope agreement	Must happen first. Everything commercial is blocked until pricing and compliance validation are confirmed.
Phase B Commercial Build	Feature-plan segmentation Payment gateway integration Subscription/billing system Final pricing on marketing pages	Blocked by: Phase A pricing agreement and feature tier decisions.
Phase C Legal & Compliance	Terms of Service Privacy Policy SaaS Agreement Data processing documentation	Can run in parallel with Phase B. Should be complete before public launch.
Phase D Pre-Launch	Expanded AI features Email marketing setup Beta user onboarding QA stress testing	Can be partly phased post-launch. AI features depend on agreed scope from Meeting #2.
LAUNCH June 2026	Public availability First paying customers Monitor and iterate	Target date. Phases A, B, and C must be complete. Phase D items that are not ready can launch as a Phase 2 roadmap.